

BACKEND DEVELOPMENT

AI StudyBuddy – Personalized Learning Assistant

1. Backend Overview

The backend manages user accounts, study materials, AI conversations, quizzes, study plans, flashcards, and student progress. It provides REST APIs connecting the frontend, database, and AI service.

2. Technologies Used

- Python
- Flask
- MongoDB
- Google Gemini API
- PyMuPDF
- REST API
- .env environment configuration

3. Backend Structure

```
backend/  
■■■ app.py  
■■■ config.py  
■■■ database.py  
■■■ requirements.txt  
■■■ .env  
■■■ models/  
■■■ routes/  
■■■ services/
```

4. Database Connection

```
from pymongo import MongoClient  
  
client = MongoClient("mongodb://localhost:27017/")  
db = client["AI_StudyBuddy"]  
students = db["students"]  
study_materials = db["study_materials"]  
quizzes = db["quizzes"]  
progress = db["progress"]
```

5. Flask Application

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route("/")  
def home():  
    return {"message": "AI StudyBuddy Backend Running"}  
  
if __name__ == "__main__":  
    app.run(debug=True)
```

6. AI Chat API

Student Question → Frontend → Flask API

7. Important REST APIs

Method	Endpoint	Purpose
POST	/api/register	Register student
POST	/api/login	Student login
POST	/api/chat	Ask AI tutor
POST	/api/material/upload	Upload material
GET	/api/materials	View materials
POST	/api/quiz/generate	Generate quiz
POST	/api/quiz/submit	Submit quiz
GET	/api/progress	View progress
POST	/api/study-plan	Create study plan
GET	/api/flashcards	View flashcards

8. Backend Workflow

Login → Authentication → Dashboard → Upload Material → PDF Processing → AI Analysis → Generate Notes/Quiz → Save Results → Track Progress

9. Security

- Password hashing
- API keys in environment variables
- Input validation
- Authentication and authorization
- Secure file-upload validation
- Error handling
- MongoDB access controls

10. Expected Result

The backend provides a secure API layer between the AI StudyBuddy frontend, MongoDB database, and AI service, enabling AI tutoring, notes, quizzes, study planning, flashcards, and progress tracking.